

Basic Concept of Function

Constant: It is a symbol which retains the same value throughout a set of mathematical operations. Constant are of two types: Absolute and arbitrary.

Constant of the type 15, 7, -100 etc. do not change whatever we may perform are known as absolute constants whereas m and c in the equation $y = mx + c$ of a straight line are arbitrary constants as these remain fixed for a particular straight line but vary from straight line to straight line.

Variable: A variable is a quantity which is capable of taking various numerical values. Variables are of two types: Dependent and Independent.

Consider the equation $y = \log_e x$ in which x can take any value, but for each value of x , there exist a value of y . Here x is called independent variable and y whose value depends upon that of x is called dependent variable.

Relation:

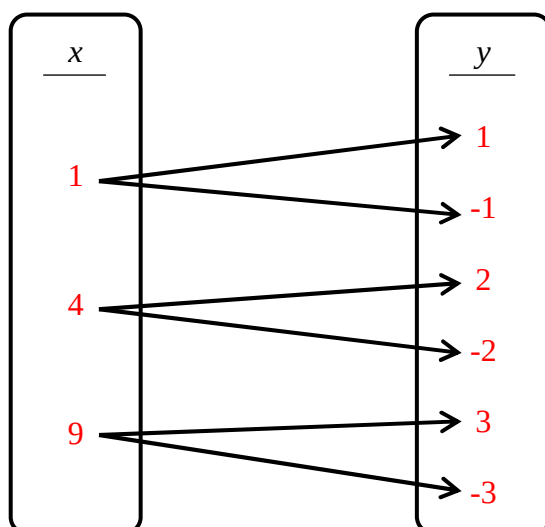
Let A and B be two non-empty sets. If each element of A is related to one or more elements of B , then it is called a Relation.

Example:

$y^2 = x$ is a relation.

For, $x = 1, 4, 9, \dots$ we get $y = \pm 1, \pm 2, \pm 3, \dots$

i.e. $(x, y) = (1, 1), (1, -1), (4, 2), (4, -2), (9, 3), (9, -3), \dots$



Function:

Let A and B be two non-empty sets. If each element of A is related to unique element of B , then it is called a function from the set A to B . It is defined by, $f : A \rightarrow B$.

Domain: The set A is called the domain of the function.

Range: The set of the images of the elements of the set A is called the range of the function.

Example:

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$y = x^2$ is a function.

For, $x = \pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \dots$ we get $y = 1, 4, 9, 16, 25, \dots$

i.e. $(x, y) = (1, 1), (-1, 1), (2, 4), (-2, 4), (3, 9), (-3, 9), (4, 16), (-4, 16), (5, 25), (-5, 25), \dots$

Since we get unique y for x , so it is a function.

Note:

1. $y^2 = x$ is not a function since we do not get unique y for x . But we have observed that $y^2 = x$ is a relation.
2. Every function is a relation but every relation is not a function.

One-one function (Injective):

Let $f : A \rightarrow B$ be a function. If distinct element of A has distinct images in B then the function is called one-one function.

Onto function (Surjective):

Let $f : A \rightarrow B$ be a function. If all elements of the set B are images of the elements of A then the function is called onto function.

One-to-one correspondence (Bijection):

The function $f : A \rightarrow B$ is a one to one correspondence, if it is both one-to-one and onto.

Example:

Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$ and $f(d) = 3$.

Is f an onto or one - one function?

Solution:

Since all three elements of the codomain are images of elements in the domain we see that f is onto. But f does not take on different values at the four elements of its domain. a and d both have same image 3. So f is not one-one.

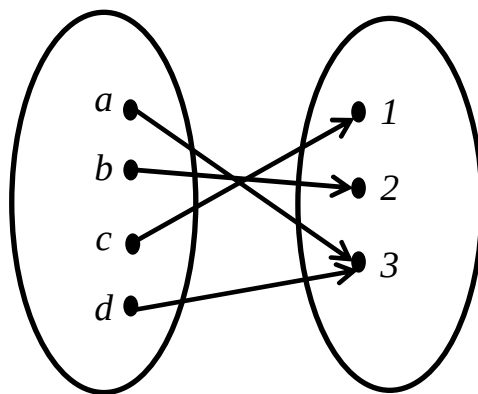


Figure: An onto function but not one-one

Some figures to distinguish different types of function:

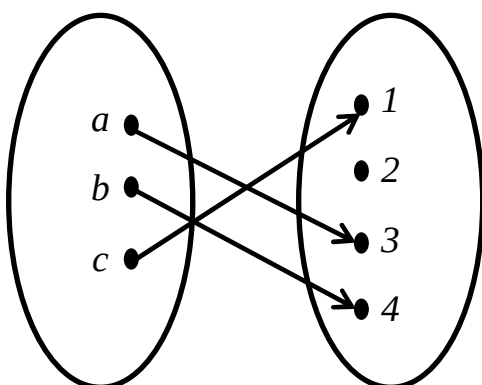


Figure (a): One-one but not onto

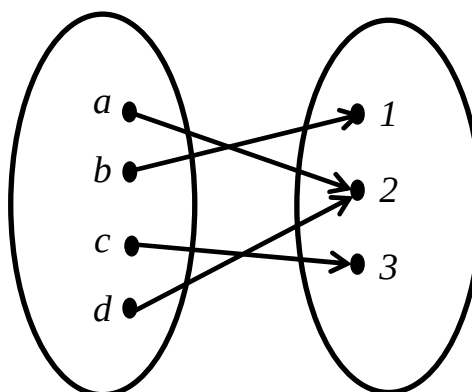


Figure (b): Onto but not one-one

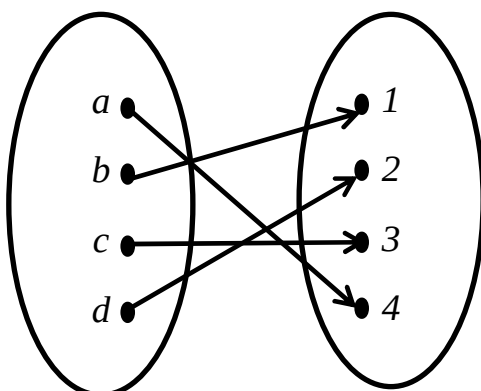


Figure (c): One-one and onto

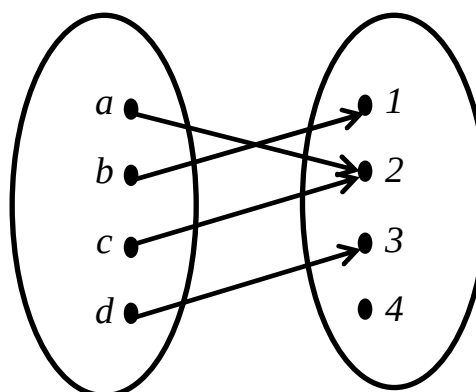


Figure (d): Neither one-one nor onto

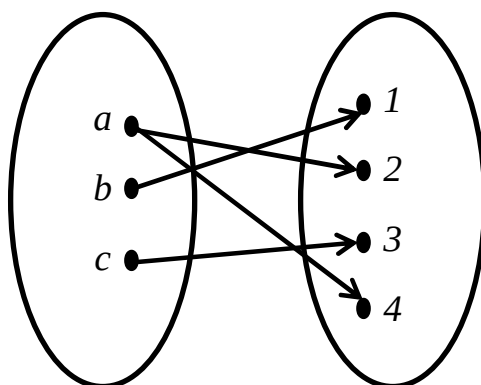


Figure (e): Not a function